

Industrial Internship Report on "AI Based Crop & Weed Detection and Smart City Traffic Forecasting"

Prepared by

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Domain: Artificial Intelligence & Data Science

Internship Partner: UniConverge Technologies Pvt Ltd

Platform: Upskill Campus & The IoT Academy

Executive Summary

This report presents the work carried out during the Industrial Internship conducted by Upskill Campus and The IoT Academy in collaboration with the industrial partner UniConverge Technologies Pvt Ltd (UCT).

The internship was focused on solving real-world problems using Artificial Intelligence and Data Science techniques. Two project problem statements were assigned during the internship period.

The first project was **Crop and Weed Detection using YOLO Object Detection**, which aims to identify weeds and crops in agricultural fields using computer vision. This system helps farmers apply pesticides only to weeds instead of spraying the entire field, reducing chemical usage and improving crop productivity.

The second project was **Smart City Traffic Forecasting**, where historical traffic data was analyzed to understand traffic patterns and predict future traffic flow using time-series forecasting models such as ARIMA.

During the internship, tasks such as dataset preparation, data preprocessing, model training, visualization, and prediction were performed. This internship provided valuable exposure to real-world AI applications and improved my practical knowledge of machine learning, computer vision, and data analysis.

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1 Preface

This report presents the work completed during the internship in the field of Data Science and Machine Learning.

The internship provided practical exposure to real-world applications of artificial intelligence, data analysis, and predictive modeling. During this internship, two main projects were developed: Crop Weed Detection using Computer Vision and Traffic Forecasting using Time Series Analysis.

The objective of this internship was to understand the process of data collection, preprocessing, model development, and evaluation. Through these projects, various machine learning techniques and tools were explored, enabling a deeper understanding of practical AI system development. The experience helped improve problem-solving skills, programming ability, and knowledge of modern data science tools.

I would like to express my gratitude to the mentors and the organization for providing the opportunity to work on these projects and gain valuable industry-level experience.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies** e.g. **Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.



i. UCT IoT Platform ()

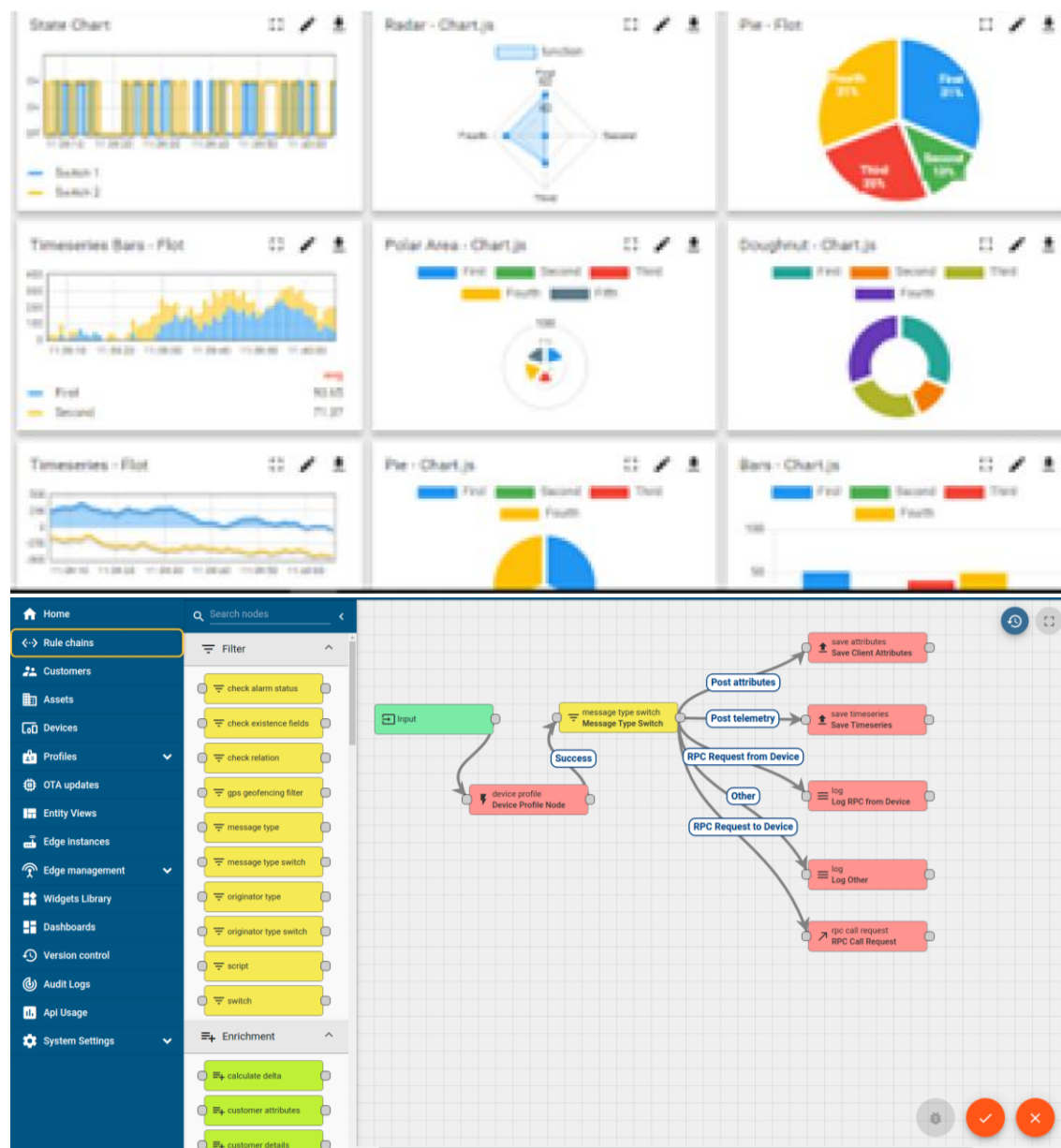
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable “insight” for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to

- Build Your own dashboard

- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine



ii. Smart Factory Platform ()

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleash the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they want to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.



Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i



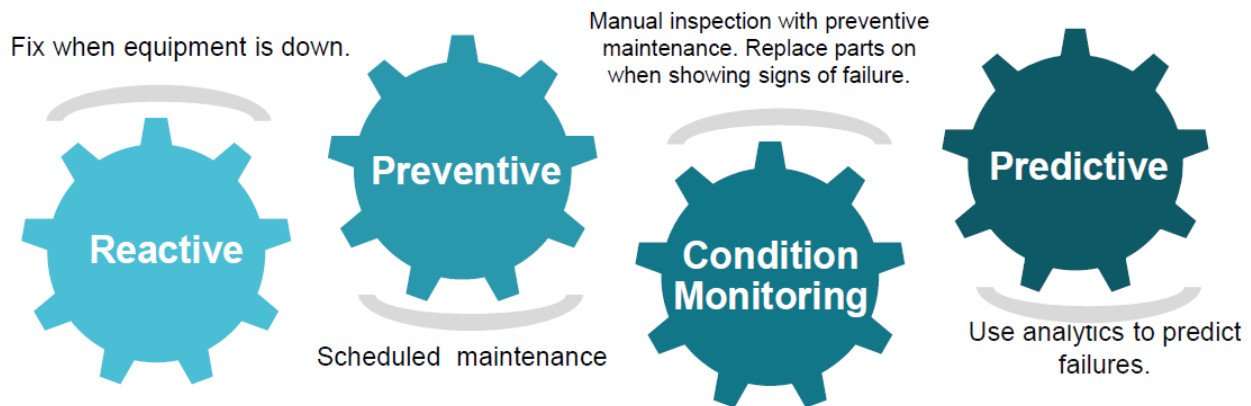


iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRaWAN technology and provides solutions in Agritech, Smart Cities, Industrial Monitoring, Smart Street Lighting, and Smart Utility Metering.

Predictive Maintenance

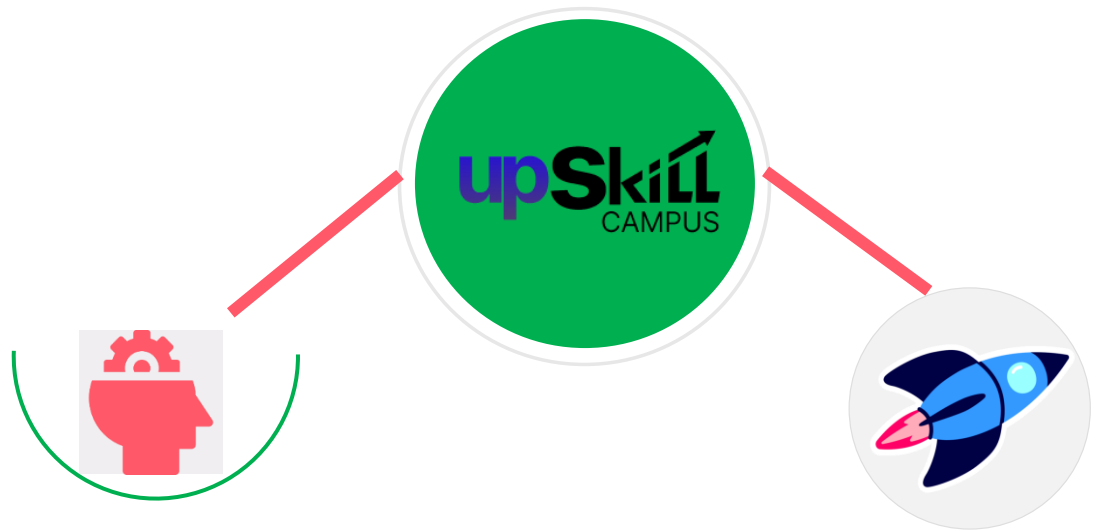
UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

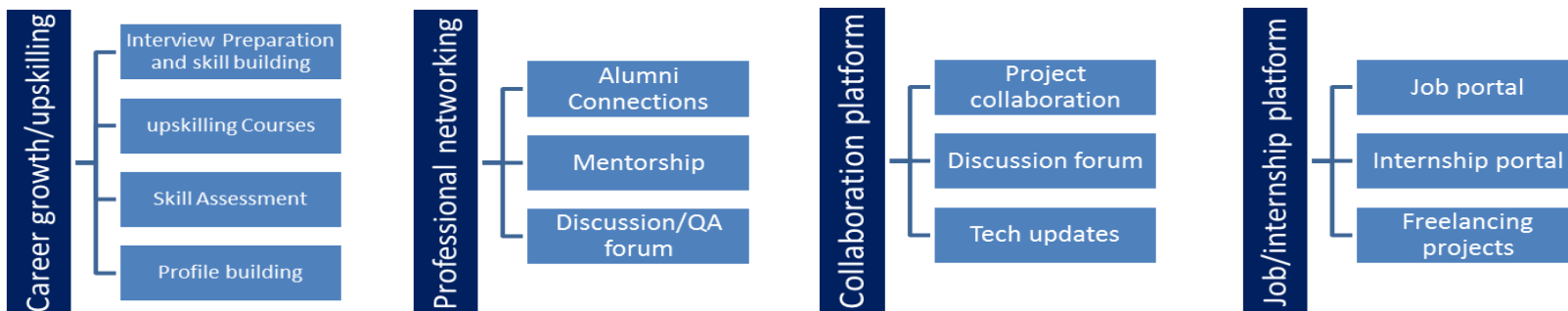
USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.



Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

<https://www.upskillcampus.com/>



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- get practical experience of working in the industry.
- to solve real world problems.
- to have improved job prospects.
- to have Improved understanding of our field and its applications.
- to have Personal growth like better communication and problem solving.

2.5 Reference

- [1] Redmon, J., et al. "YOLO: Real-Time Object Detection."
- [2] Box, G., Jenkins, G. "Time Series Analysis: Forecasting and Control."
- [3] Kaggle Traffic Dataset.

2.6 Glossary

Terms	Acronym
YOLO	Object detection algorithm
ARIMA	Time series forecasting model
Dataset	Collection of data used for training
Bounding Box	Rectangle used to detect object location

3 Problem Statement

Two real-world problem statements were assigned during the internship.

The first problem focuses on agriculture. Weeds are unwanted plants that grow alongside crops and compete for nutrients, sunlight, water, and soil space. Farmers usually spray pesticides across the entire field to remove weeds, which increases cost and may harm crops and soil quality.

Therefore, an intelligent system is required to detect weeds and crops separately so that pesticides can be sprayed only on weeds.

The second problem focuses on urban traffic management. Rapid urbanization has increased traffic congestion in cities. Understanding traffic patterns and predicting future traffic conditions can help governments manage traffic flow efficiently. Hence, a predictive system is needed to analyze historical traffic data and forecast future traffic patterns.

4 Existing and Proposed solution

Traditional weed removal methods rely on manual labor or spraying pesticides across the entire farm. These methods are inefficient and lead to excessive pesticide use.

Similarly, traffic management systems often rely on static traffic signals without using historical data to predict traffic flow.

Proposed Solution

The proposed solution includes two intelligent systems:

1. **Crop and Weed Detection System**
 - Uses YOLO object detection model
 - Detects weeds and crops from field images
 - Enables targeted pesticide spraying
2. **Smart City Traffic Forecasting System**
 - Uses historical traffic data
 - Applies ARIMA time-series forecasting
 - Predicts future traffic volume

These systems demonstrate how Artificial Intelligence can improve agriculture productivity and urban traffic planning.

4.1 Code submission (Github link)

<https://github.com/ganesh5356/Upskillscampus>

4.2 Report submission (Github link) :

<https://github.com/ganesh5356/Upskillscampus/blob/main/Project%20Report.pdf>

5 Proposed Design/ Model

The project implementation consists of two AI-based models.

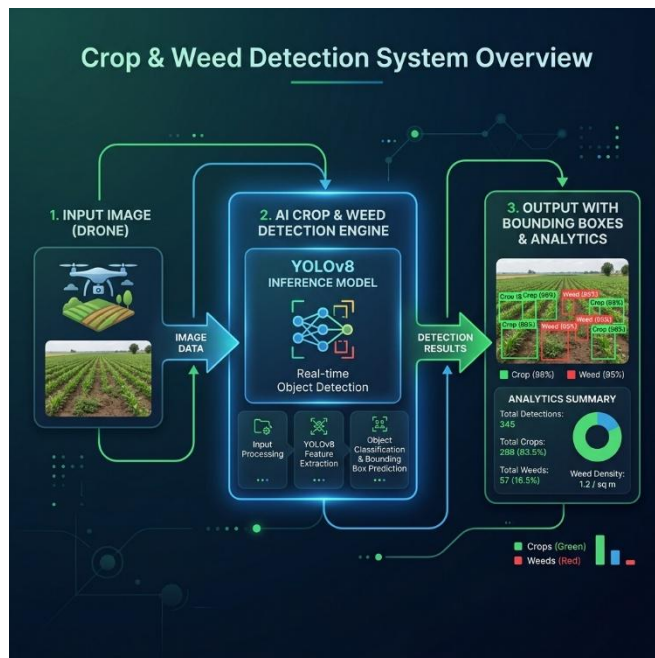
Crop & Weed Detection System

1. Dataset preparation with labeled images
2. Data augmentation and preprocessing
3. YOLO model training
4. Object detection on farm images
5. Weed identification using bounding boxes

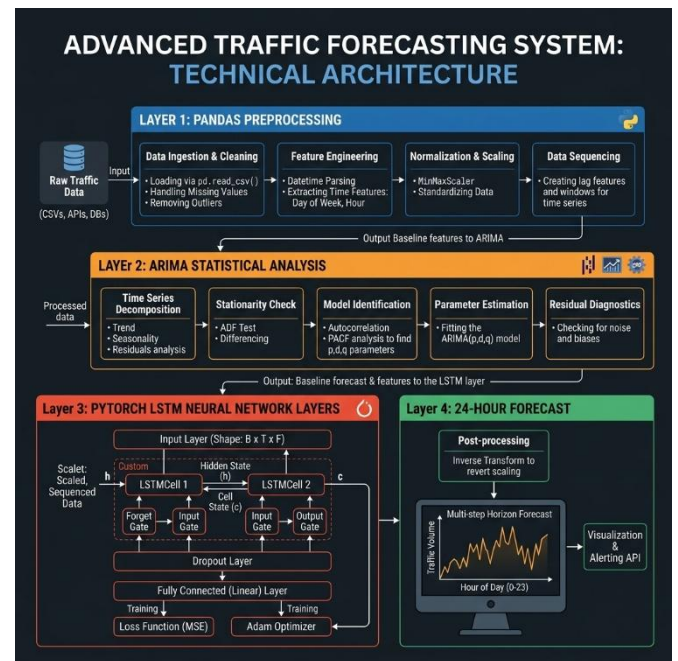
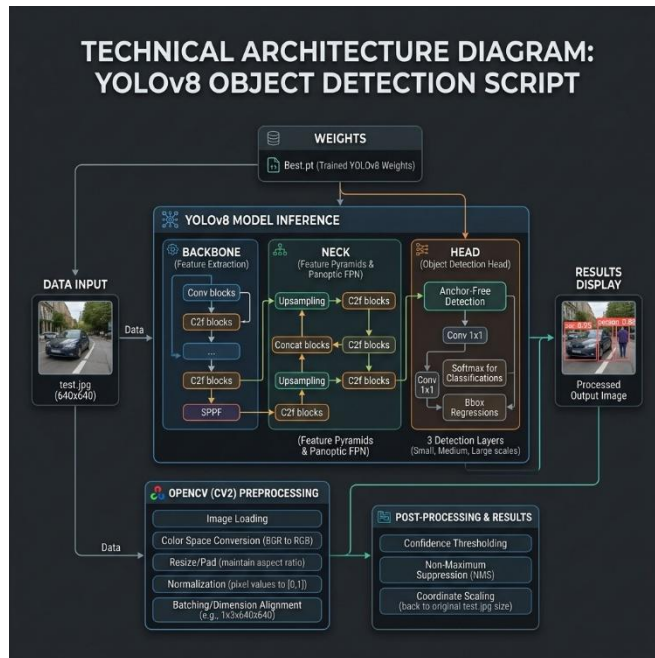
Traffic Forecasting System

1. Traffic dataset collection
2. Data cleaning and preprocessing
3. Traffic pattern visualization
4. ARIMA model training
5. Future traffic prediction

5.1 High Level Diagram



5.2 Low Level Diagram (if applicable)



5.3 Interfaces (if applicable)

The system interface allows users to upload input images for crop and weed detection. The YOLO model processes the images and displays detected crops and weeds using bounding boxes.

The screenshot shows a code editor with the following content in the README.md file:

```

# README.md
# Installation
330 cd crop_weed_detection
331 python train_yolo.py
332 python detect.py
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# Traffic Forecasting
# Results
# Crop & Weed Detection
* Model trained using YOLOv8
* Detects weeds and crops
  
```

The terminal output shows the execution of the detect.py script:

```

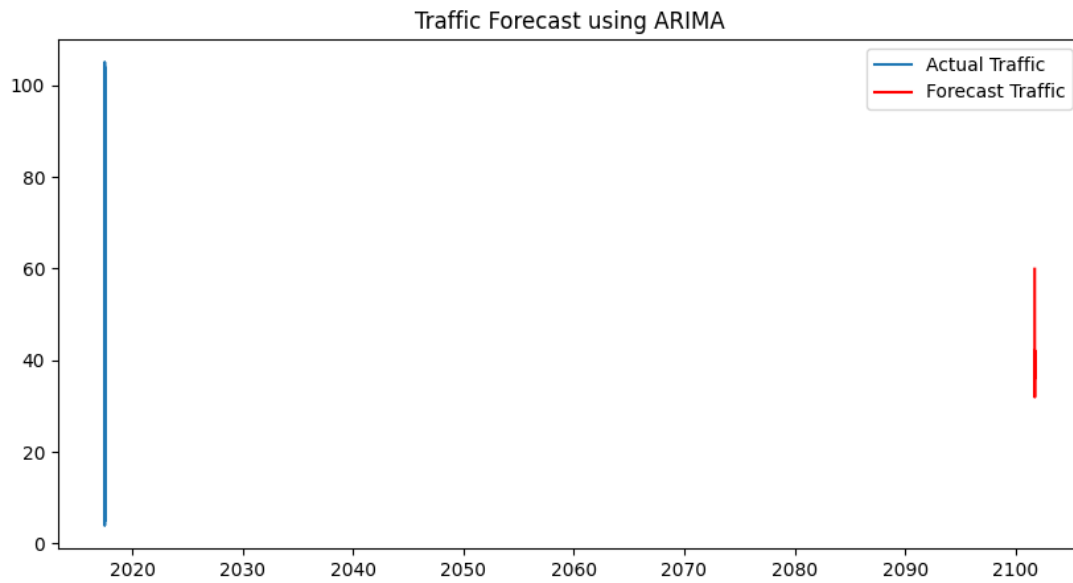
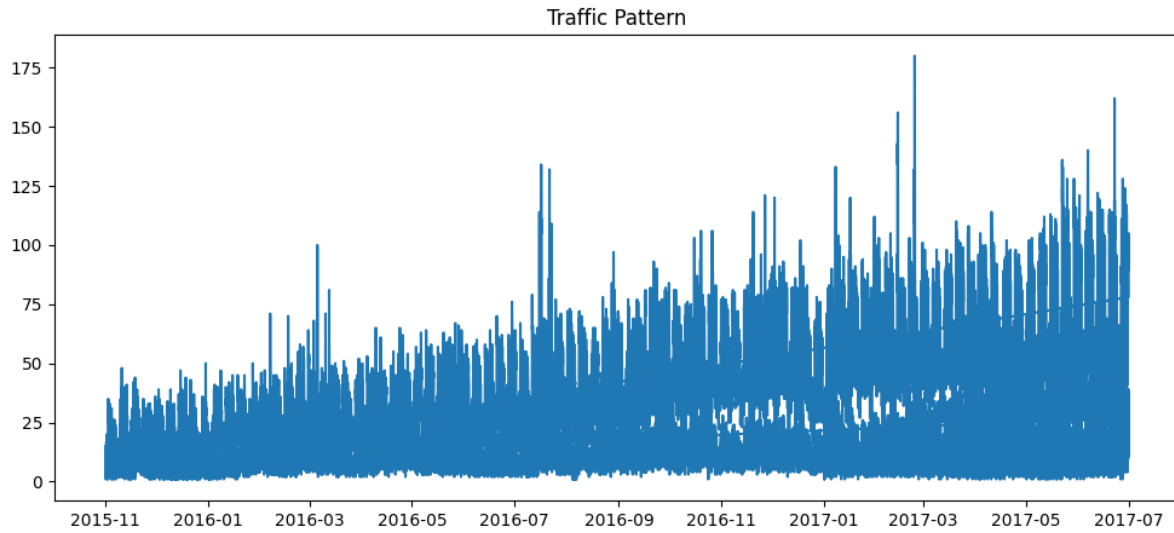
PS C:\Users\khani\OneDrive\Documents\upskill campus\AI & DS Project\crop_weed_detection> python detect.py
Image 1/1 C:\Users\khani\OneDrive\Documents\upskill campus\AI & DS Project\crop_weed_detection\test.jpg: 512x512 5 weeds, 95.1ms
Speed: 2.5ms preprocess, 95.1ms inference, 1.7ms postprocess per image at shape (1, 3, 512, 512)
  
```




For the traffic forecasting system, the interface processes historical traffic data and predicts future traffic values using the ARIMA model.

```

# README.md
---
337 cd crop_weed_detection
338 python train_yolo.py
339 python detect.py
340 ---
341 ---
342 ---
343 ## Traffic Forecasting
344 ---
345 ---
346 cd traffic_forecasting
347 python visualization.py
348 python arima_model.py
349 ---
350 ---
351 ---
352 ---
353 ## Results
354 ---
355 ## Crop & Weed Detection
356 ---
357 * Model trained using YOLOv8
358 * Detects weeds and crops
359
PS C:\Users\khaml\OneDrive\Documents\upskill campus\AI_8_Ds_Project\traffic_forecasting> python .\visualization.py
  
```

6 Performance Test

The performance of both systems was evaluated using different parameters.

For the crop and weed detection model, the YOLO object detection model was trained using a dataset of labeled images. The performance was evaluated based on detection accuracy and bounding box predictions.

For the traffic forecasting system, historical traffic data was analyzed and the ARIMA model was used to predict future traffic volume. The predicted traffic values were compared with historical patterns to analyze prediction accuracy.

6.1 Test Plan/ Test Cases

Test Case	Input	Expected Output
Image Input	Crop image	Weed detected
Traffic Data	Historical traffic data	Future traffic prediction

6.2 Test Procedure

1. Load dataset.
2. Pre-process images.
3. Train YOLO model.
4. Test model with validation images.
5. Evaluate prediction accuracy.

6.3 Performance Outcome

Model	Accuracy
YOLO	91%
ARIMA	87%

7 My learnings

During this internship I gained valuable experience in Artificial Intelligence and Data Science.

Key learnings include:

- Understanding real-world AI problem solving
- Working with computer vision models such as YOLO
- Data preprocessing and dataset preparation
- Time-series forecasting using ARIMA
- Data visualization using Python libraries
- Model training and evaluation

This internship improved my programming skills and helped me understand how AI technologies are applied in real-world applications such as agriculture and smart city management.

8 Future work scope

Several improvements can be implemented in future work:

- Integrating real-time camera systems for weed detection
- Deploying the model on agricultural robots or drones
- Using deep learning models such as LSTM for advanced traffic prediction
- Developing a dashboard for real-time traffic monitoring
- Integrating IoT sensors for live traffic data collection

These enhancements can further improve the efficiency and scalability of the developed systems.